

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	LGT5073
Subject Title	Risk Management in Operations
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	None ISE548 Risk and Crisis Management
Objectives	This subject seeks to develop the knowledge and analytical/practical skills necessary in organizations, with strong emphasis on operations management and quality management, for making risk management decisions to ensure business growth and survival through the application of the principles and practices of the full spectrum of risk management framework. This subject contributes to the following Intended Learning Outcomes for MSc in Operations Management: • #2: Develop the specific operations management knowledge
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Analyse the inherent risks in businesses and operations by applying the correct understanding and basic principles of risk and risk management. - Use the correct risk management concepts to devise appropriate strategies and tactics for risk management. - Comprehend the logical approaches and sequential steps of risk management comprising definition of objective, identification, assessment (analysis and measurement), selection of risk management strategies, implementation of appropriate risk management actions, and finally measurement and evaluation of the performance of the risk management actions. - Master the risk management tools and techniques for financial risks, particularly the typical price risks. - Be familiar with risk management in operations to a level that is adequate for continued self-enhancement of knowledge and practical applications of risk management.

	f. Explore and understand how emerging technologies (for examples, artificial intelligence, blockchain, cloud computing etc.) lead to emerging risks that cause negative and positive impacts on business objectives, and how the emerging risks are managed and mitigated by risk management.
Subject Synopsis/ Indicative Syllabus	Introduction and Understanding the Correct Principles and Concepts of Risks: origin of risk, definition of risk, elements of risk, risk and uncertainty, risk perception, risk exposure, risk response, classification of risk, sources of risk, causes of risk, typical organizational risks in businesses and operations, and supply chain risks. Fundamental of Risk Management: development of risk management, understanding of risk management, contributions of risk management, roles and responsibilities for risk management, and enterprise of risk management. Risk Management Process for Negative Risks: the logical and sequential steps of risk management process covering determination of risk management objectives (aligns with corporate objective), identification of all potential and inherent risks, assessment and evaluation of risks (including risk analysis, risk measurement and the use of risk matrix), selection of risk management strategies, identification of risk management actions (the logical steps to identify risk problems and root causes (risk factors and hazards) on the basis to determine the risk management actions), implementation of risk management actions, and finally the performance measurement of the effectiveness and efficiency of risk management actions. Risk Management Strategies and Techniques: risk management strategies for negative and positive risks, the corresponding techniques to manage negative risks. Tools and Techniques for Managing Financial Risks: the use of four main types of derivatives to hedge and manage speculative risks. Risk Culture: national culture and organizational culture, chain effect of culture, overview of organizational culture and its determinants, risk and organizational culture, risk culture, and revisit of risk perception and risk attitude. Supply Chain Risk and Risk Management: fundamental of supply chain risks, overview and understanding of supply chain risk management, and supply chain risk management process. Risk Management Framework and Emerging Technologies: explore the negative and positive risks (technology risks) arising from emerging technologies (for examples, artificial intelligence, blockchain, cloud computing etc.) and these risks can be managed effectively and efficiently in the digital transformation era.

Teaching/Learning Methodology	Lecture: Learn academic concepts and practical approaches/tools/techniques/ methods of the risk management framework aims to allow students to acquire the correct understanding of the principles and concepts of risk and risk management, and then putting and applying the academic concepts and practical applications of risk management approaches, tools, techniques, and methods into the real-life operational, commercial, and business contexts. Coursework and final examination: Learn to practically apply risk management approaches, tools, techniques and methods, and to study selected topics in-depth.																																																														
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Class attendance ^(Note 1)</td><td>5%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Individual assignment ^(Note 2)</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Case study ^(Note 3) (simulation presentation + individual assessment)</td><td>20%</td><td></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Group project ^(Note 4)</td><td>35%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Quiz ^(Note 5)</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The coursework focuses on both knowledge acquisition and practical application of risk management through individual and group efforts. Learning from practical and work-based experiences forms an important component of students’ learning and assessment. The coursework of this subject is designed to assess the students’ abilities to understand, comprehend, master, and apply the knowledge, skills, techniques, and methods within the risk management framework. It reinforces the academic concepts and practical applications learned during the lectures, enabling their utilization in real-life operational, commercial, and business contexts. Note 1: The attendance (5%) shall encourage students to attend ad participate in classes in interactive teaching and learning methods as provided in ‘Teaching/Learning Methodology’ above. Note 2: There is a large pool of research knowledge on risk management and its relevant areas. Thirty articles on risk management topics have been specifically selected for the individual assignment (20%). Each student	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	Class attendance ^(Note 1)	5%	✓	✓	✓	✓	✓	✓	Individual assignment ^(Note 2)	20%	✓	✓	✓	✓	✓	✓	Case study ^(Note 3) (simulation presentation + individual assessment)	20%		✓	✓	✓	✓	✓	Group project ^(Note 4)	35%	✓	✓	✓	✓	✓	✓	Quiz ^(Note 5)	20%	✓	✓	✓	✓	✓	✓	Total	100 %						
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	is assigned one article to prepare an academic write-up covering (i) what the student has learned from the article, (ii) how the study relates to what we have discussed in classes, (iii) how the article reinforces the student's learning of risk management, and (iv) critiques of the article. The individual assignment is designed and utilized to help students enhance their knowledge in risk management to achieve the intended subject learning outcomes (a – f). Note 3: Students in a group are assigned one of the three prepared case studies in specific risk management areas. Students are required to prepare and present their analyses and findings for the assigned case study (20%) in class, followed by a Q&A session. Suggested answers will be provided and discussed after the presentations. The case study aims to enhance the students' knowledge and abilities in handling the three specific risk management areas to the intended subject learning outcomes (b – f). Note 4: Students in a group are required to prepare and present a real-life (case-driven) or hypothetical (topic-driven) group project (35%) on risk management focusing on a project, operation, or business organization. The students shall demonstrate their abilities to prepare and deliver a commercial presentation in business settings related to risk management. The group project assess the students' abilities to strengthen their practicable knowledge, skills, and application of risk management, aiming to achieve the intended subject learning outcomes (a – f) with an emphasis on (c – f). Note 5: The quiz (20%) tests the students' abilities to understand and comprehend the basic and fundamental approaches tools, techniques, and methods of the risk management framework, aiming to achieve the intended learning outcomes (a – f). To reflect the significant technology content in this subject, 10% (or more) of the overall weighting of this subject is based on individual assessment concerning technology-related knowledge.	
Student Study Effort Expected	Class contact:	
	Lectures / tutorials (if any)	39 hrs.
	Other student study effort:	
	Self-study for preparing lectures and tutorials (if any)	26 hrs.
	Preparation of coursework	61 hrs.
	Total student study effort	126 hrs.
Reading List and References	<u>Recommended Reference Books:</u> • Blunden, T & Thirlwell, J. (2010). <i>Mastering operational risk</i>. Harlow, England, New York: Financial Times Prentice Hall. • Devlin, E.S. (2007) <i>Crisis management planning and execution</i>. Boca Raton, FL: Auerbach Publications, c2007.	

	• Haimes, Y. Y. (2004) <i>Risk Modeling, Assessment and Management</i>. New York: Wiley. • Handfield, R.B. & McCormack, K. (ed.) (2008) <i>Supply chain risk management: Minimizing disruptions in global sourcing</i>. Boca Raton, Fla.: Auerbach Publications. • Hubbard, D.W. (2009) <i>The failure of risk management: Why it's broken and how to fix it</i>. Hoboken, N.J.: J. Wiley & Sons. • Oliver, E.C. (2011). Catastrophic disaster planning and response [electronic resource]. Boca Raton: CRC Press. • Trim, P.R.J. & Caravelli, J. (ed.) (2009). <i>Strategizing resilience and reducing vulnerability</i>. New York: Nova Science Publishers. <u>Recommended Journals:</u> • International Journal of Risk Assessment and Management • Journal of Risk and Financial Management • Risk Management in Financial Institutions • Journal of Business Continuity & Emergency Planning • Institute of Risk Management (IRM) • The Public Risk Management Association, US (PRIMA) • The Public Risk Management Association, UK (ALARM) • Association of Insurance and Risk Managers <u>Recommended International Standards and Frameworks:</u> • ISO 3100:2018 for Risk Management • ISO/IEC 27005:2018 for Information Security System • ISO 22301:2019 for Business Continuity Management (focusing on the resilience of organizations to disruptions) • COSO ERM Framework • PMBOK Guide for Project Management
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